

Cloudinary Media Handling

Overview

Cloudinary is the primary media delivery path for uploaded images and videos. The backend also rewrites outgoing Cloudinary URLs to improve delivery size and quality automatically.

Current implementation

Upload configuration

- file: `config/cloudinary.js`
- configured from:
 - `CLOUDINARY\_CLOUD\_NAME`
 - `CLOUDINARY\_API\_KEY`
 - `CLOUDINARY\_API\_SECRET`
- profile uploads default to folder `yenkasa/profile`

Response optimization

- file: `utils/cloudinaryMedia.js`
- middleware: `cloudinaryMediaResponseOptimizer`
- rewrites JSON responses containing Cloudinary delivery URLs
- injects:
 - `f\_auto`
 - `q\_auto`
 - width/crop transforms
- generates a poster image for videos when one is missing

Important characteristics

- media optimization is applied centrally at `res.json(...)`
- large uploads are audit-logged
- Cloudinary handling is deeply integrated into API response shaping

Known issues

- response mutation at the framework level can make debugging payloads harder
- media transformations are inferred from field names, which is useful but heuristic

Scaling concerns

- global JSON response rewriting adds work to every matching API response
- transformation rules are code-driven rather than policy-configured

Recommended improvements

1. document canonical media field semantics
2. precompute more media derivatives during upload if needed
3. expose media optimization rules in a dedicated configuration module